

WASP-12b

R-1021

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_161230 · created 2026-07-30T04:46:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457752.67613 +/- 0.00035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.045
Transit depth (Rp/Rs)^2	2.36 +/- 1.37 [%]
Orbital Inclination (inc)	76.95 +/- 2.88
Airmass coefficient 1 (a1)	1.05 +/- 0.95
Airmass coefficient 2 (a2)	0.14 +/- 0.21
Scatter in the residuals of the lightcurve fit is	114.09 %
Best Comparison Star	#1 - [26, 312]
Optimal Aperture	1.94
Optimal Annulus	5.3
Transit Duration (day)	0.108 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.045 vs 0.1178 (deviation 0.59σ)
Duration fit vs published	0.108 d vs 0.125 d (deviation 0.8σ)
Mid-time O–C	-0.9 min
Depth SNR	2.5
Residual scatter	114.09 %

Light curve

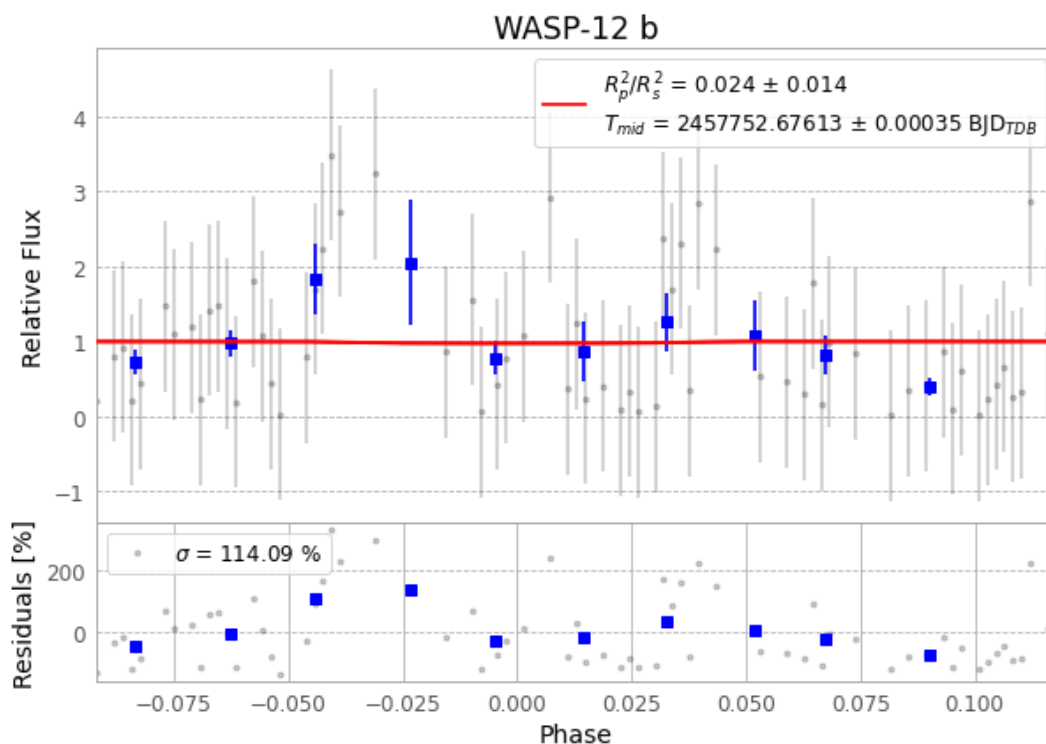


Figure 1 — FinalLightCurve_WASP-12 b_2016-12-29.png (SHA-256 ef84cbd5db692dc4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [74, 166], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 709547b2a09202aa...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 902034c

Data provenance

114 FITS frames (75 MB), WASP-12161230013612.FITS → WASP-12161230071512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-161230.log (bfeb9f11cdd7...), FinalLightCurve_WASP-12 b_2016-12-29.png (ef84cbd5db69...), FinalLightCurve_WASP-12 b_2016-12-29.csv (c882b14c1d34...)

Compute resources

Wall time 265.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 04:46Z.